# Atomic Detection Pipeline

MITRE ATT&CK Technique Emulation & SIEM Detection Report

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## 1. Project Overview

This report documents the successful implementation of a threat detection pipeline using free, open-source security tools. The objective was to emulate a real-world MITRE ATT&CK technique on a Windows victim machine and prove that a SIEM (Wazuh) detected and alerted on it.

## 2. Lab Environment

All machines were run as VirtualBox virtual machines on a single host. The network was configured to allow communication between the Windows victim and the Wazuh manager.

| Component | Tool / Version | Role |
|---|---|---|
| SIEM / Manager | Wazuh v4.14.3 OVA | Collects logs, applies detection rules, generates alerts |
| Windows Victim | Windows 10 (x64) | Target machine where the attack technique was emulated |
| Attack Emulation | Atomic Red Team v2.1.0 | Executes MITRE ATT&CK mapped test cases |
| Telemetry | Sysmon (SwiftOnSecurity config) | Deep Windows event logging (process, network, file) |
| Wazuh Agent | ossec-agent (Wazuh 4.x) | Ships Sysmon & Windows logs to the Wazuh manager |

## 3. Pipeline Architecture

The detection pipeline follows a linear flow from attack emulation through telemetry capture to SIEM alerting:

```
Atomic Red Team → Sysmon (Event Capture) → Wazuh Agent → Wazuh Manager → Alert
```

Each component plays a distinct role:

- Atomic Red Team emulates the attacker, executing a known-bad technique safely.
- Sysmon captures deep telemetry (full command lines, parent processes, hashes) that Windows Event Logs alone would miss.
- The Wazuh Agent forwards Sysmon logs from the Microsoft-Windows-Sysmon/Operational channel to the manager.
- Wazuh Manager applies detection rules, maps findings to MITRE ATT&CK, and surfaces alerts in the dashboard.

# 4. Technique Executed

## T1059.001 — Command and Scripting Interpreter: PowerShell

| Field | Detail |
|---|---|
| MITRE Technique ID | T1059.001 |
| Tactic | Execution |
| Sub-technique | PowerShell |
| Atomic Test Number | Test #1 |
| Tool Used | Invoke-AtomicTest (Atomic Red Team v2.1.0) |
| Target Machine | DESKTOP-JRU2NP1 (Windows 10) |

### What This Technique Simulates

T1059.001 covers adversary use of PowerShell to execute commands and scripts. Threat actors commonly use PowerShell for execution, lateral movement, and post-exploitation because it is built into every modern Windows system and provides access to the .NET framework and Windows APIs.

### Command Used to Execute the Test

```
Import-Module C:\AtomicRedTeam\invoke-atomicredteam\Invoke-AtomicRedTeam.psd1 -Force
Invoke-AtomicTest T1059.001 -TestNumbers 1
```

# 5. Detection Evidence

## Sysmon Event Captured

Sysmon Event ID 1 (Process Creation) captured the PowerShell execution with full command-line arguments. The ossec.conf localfile block was configured to forward the Microsoft-Windows-Sysmon/Operational channel to Wazuh:

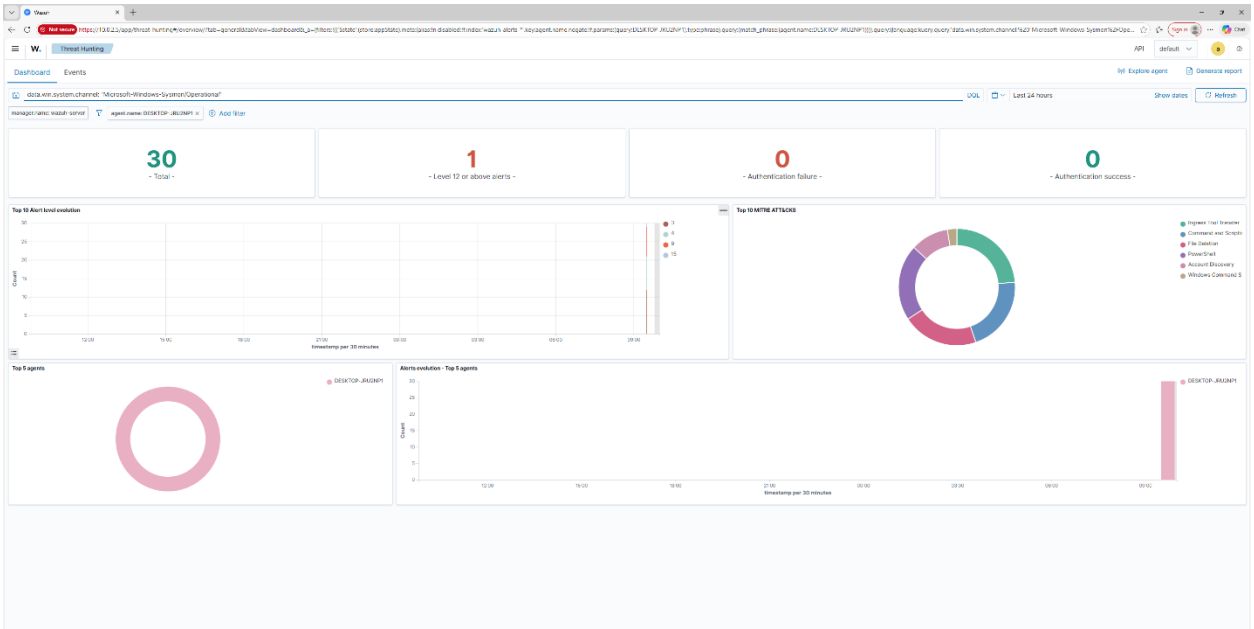
```
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

## Wazuh Dashboard Results

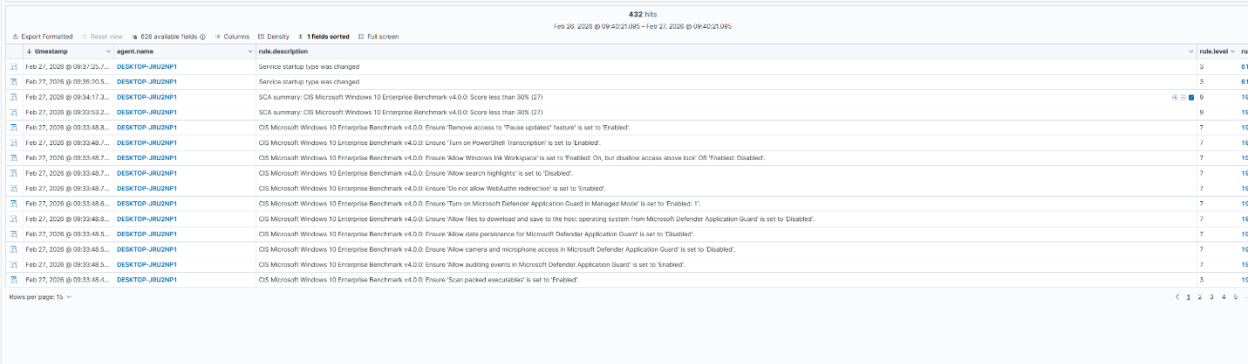
After executing the atomic test and waiting approximately 30-60 seconds for log shipping, the Wazuh Threat Hunting dashboard showed the following results filtered to the Sysmon channel:

| Metric | Value |
|---|---|
| Total Sysmon Events Detected | 30 |
| High Severity Alerts (Level 12+) | 1 |
| Agent Reporting | DESKTOP-JRU2NP1 |
| MITRE Technique Confirmed in Dashboard | PowerShell (T1059.001) |
| MITRE Tactic Confirmed | Execution |

## Screenshot 1: Wazuh Threat Hunting Dashboard — MITRE ATT&CK Wheel showing PowerShell detection




## Screenshot 2: Wazuh Events Tab — 432 security events from DESKTOP-JRU2NP1



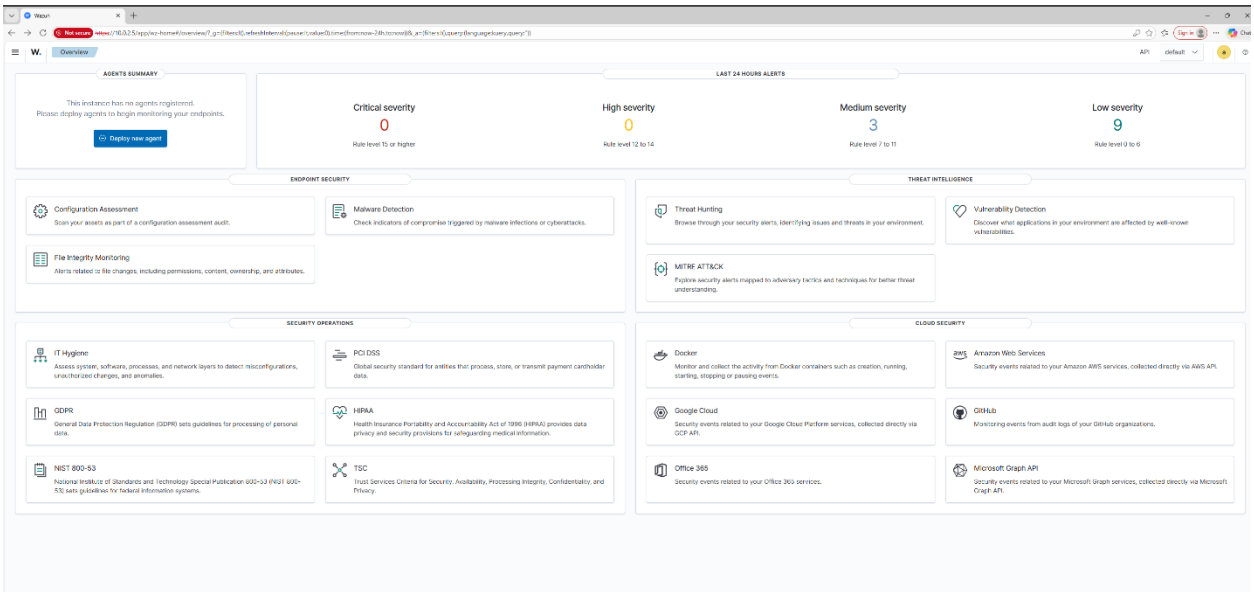
432 hits

Feb 26, 2026 @ 09:40:21.095 - Feb 27, 2026 @ 09:40:21.095

Export Formatted | Reset view | 626 available fields | Columns | Density | 1 fields sorted | Full screen

| timestamp | agent.name | rule.description | rule.level | ru |
|---|---|---|---|---|
| Feb 27, 2026 @ 09:37:25.7... | DESKTOP-JRU2NP1 | Service startup type was changed | 3 | 61 |
| Feb 27, 2026 @ 09:35:20.5... | DESKTOP-JRU2NP1 | Service startup type was changed | 3 | 61 |
| Feb 27, 2026 @ 09:34:17.3... | DESKTOP-JRU2NP1 | SCA summary: CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Score less than 30% (27) | 9 | 19 |
| Feb 27, 2026 @ 09:33:53.2... | DESKTOP-JRU2NP1 | SCA summary: CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Score less than 30% (27) | 9 | 19 |
| Feb 27, 2026 @ 09:33:48.8... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Remove access to "Pause updates" feature' is set to 'Enabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.7... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Turn on PowerShell Transcription' is set to 'Enabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.7... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Allow Windows Ink Workspace' is set to 'Enabled: On, but disallow access above lock' OR 'Enabled: Disabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.7... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Allow search highlights' is set to 'Disabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.7... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Do not allow WebAuthn redirection' is set to 'Enabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.6... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Turn on Microsoft Defender Application Guard in Managed Mode' is set to 'Enabled: 1'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.0... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Allow files to download and save to the host operating system from Microsoft Defender Application Guard' is set to 'Disabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.5... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Allow data persistence for Microsoft Defender Application Guard' is set to 'Disabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.5... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Allow camera and microphone access in Microsoft Defender Application Guard' is set to 'Disabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.5... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Allow auditing events in Microsoft Defender Application Guard' is set to 'Enabled'. | 7 | 19 |
| Feb 27, 2026 @ 09:33:48.4... | DESKTOP-JRU2NP1 | CIS Microsoft Windows 10 Enterprise Benchmark v4.0.0: Ensure 'Scan packed executables' is set to 'Enabled'. | 3 | 19 |

Rows per page: 15 | 1 2 3 4 5




## 6. Key Findings

1. The pipeline successfully detected MITRE ATT&CK technique T1059.001 end-to-end.
2. Sysmon provided the necessary process-level telemetry that basic Windows Event Logs would not have captured.
3. The Wazuh MITRE ATT&CK module correctly mapped the alert to the PowerShell execution tactic in the Threat Hunting dashboard.
4. The localfile Sysmon configuration in ossec.conf was a critical dependency — without it, Wazuh received no Sysmon data.

## 7. Conclusion

This project demonstrated a fully functional threat detection pipeline using entirely free and open-source tools. By combining Atomic Red Team for safe attack emulation, Sysmon for deep Windows telemetry, and Wazuh as the SIEM, a realistic blue team detection scenario was achieved. The pipeline successfully caught MITRE ATT&CK technique T1059.001 (PowerShell Execution) and surfaced it with proper MITRE context in the Wazuh dashboard, proving that the detection stack is operational and effective.

## 8. Project Extension: AI-Powered Alert Summarizer (Web App)

As an extension to the core detection pipeline, an AI-powered web application was developed and deployed to automate the analysis of Wazuh alert JSON. Rather than manually interpreting raw alert data, a SOC analyst can paste any Wazuh alert into the tool and receive an instant, plain-English incident summary generated by Gemini 2.0 Flash.

## Tool Overview

**Live URL:** `https://wazuhai-36zejwby.manus.space`

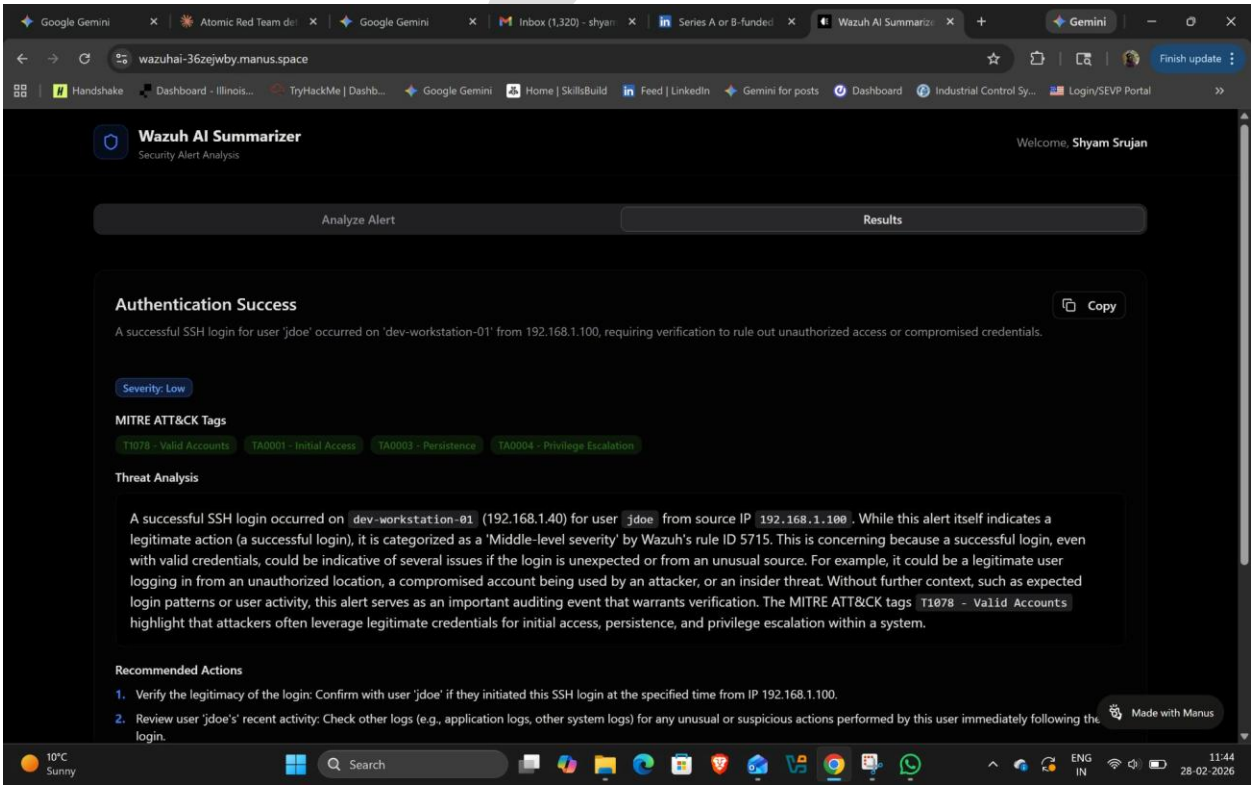
**Built with:** Manus AI (web app generation) + Google Gemini 2.0 Flash API (alert analysis)

## How It Works

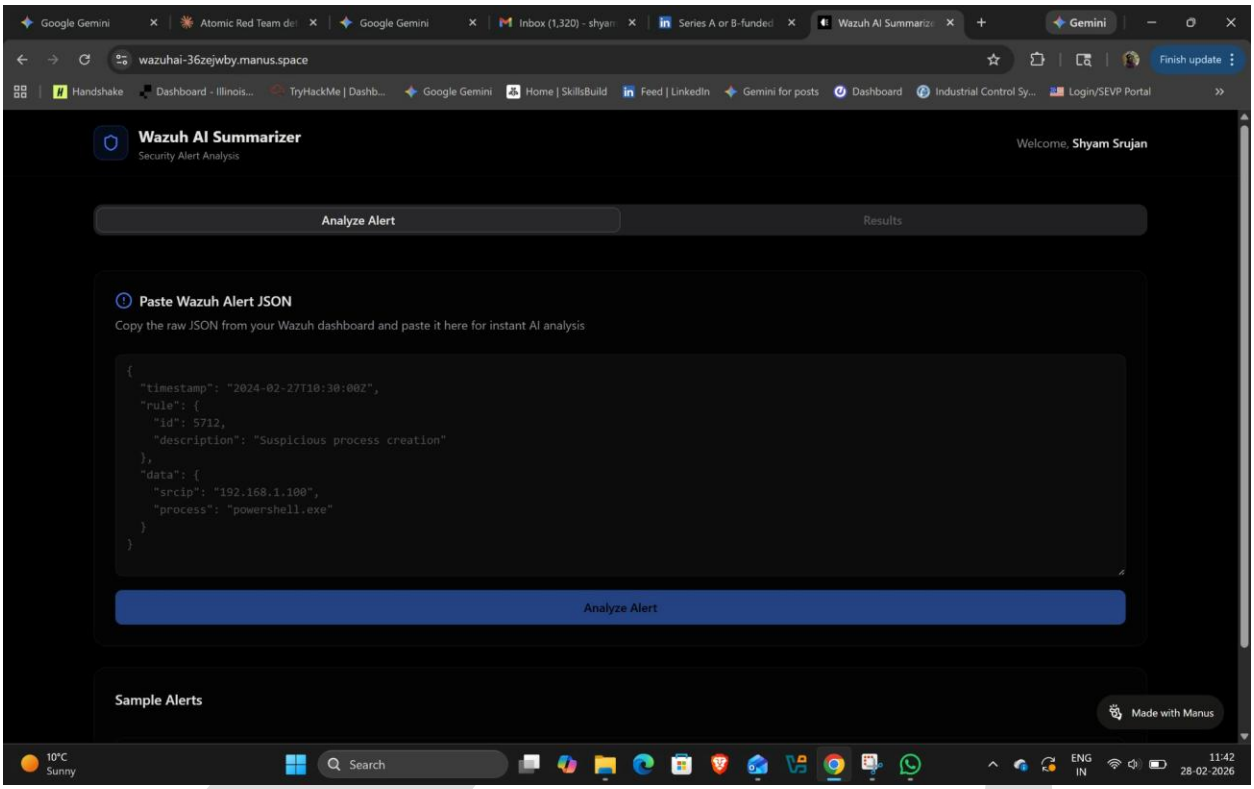
The analyst copies a raw Wazuh alert JSON from the dashboard and pastes it into the web app. Upon clicking Analyze Alert, the JSON is sent to the Gemini 2.0 Flash API which returns a structured incident report including: severity rating, MITRE ATT&CK technique and tactic tags, a plain-English threat analysis, and a numbered list of recommended response actions. The tool was built and deployed using Manus AI by describing the idea in natural language.

## Screenshots

### Screenshot 4: Alert Input — Paste Wazuh JSON for instant AI analysis




### Screenshot 5: AI Analysis Results — Severity, MITRE tags, threat analysis and recommended actions




## Value of This Extension

This extension demonstrates how AI can augment a SOC analyst's workflow. Instead of spending time manually reading and interpreting raw JSON alerts, the AI instantly produces a human-readable summary with severity, context, and actionable next steps. This mirrors capabilities found in enterprise tools such as Microsoft Copilot for Security, but built entirely with free tools and a conversational AI prompt. The extension also demonstrates practical use of the Gemini API, Manus AI for rapid app deployment, and the concept of LLM-assisted threat analysis — all relevant skills for modern cybersecurity roles.